Chapter 17 worked solutions • Distributed systems

OS17-E01 • Speedup and fault domains

1. T6=30+90/6+6=51 s; speedup=120/51≈2.353. Without coordination the infinite-node upper bound is 120/30=4.
2. At the specified instant the independent unavailability events all occur with probability $0.02^3=8\times10^{-6}$. Shared racks can correlate failures; metadata and network failures are excluded from this result.

OS17-E02 • Moving data or computation

1. Full-data network time is 512/64+0.010=8.010 s, excluding local computation. Remote execution plus result is 0.200+2/64+0.010=0.24125 s. The model charges one RTT and does not overlap computation with transfer.
2. No. Running state includes PC, registers, memory, open resources and compatible ABI; sending code alone can instead be remote execution.

OS17-E03 • BDP and windows

1. BDP=200 • 10^6 • .05=10000000 bits=1250000 B≈1220.70 KiB. Window bound=131072 • 8/.05=20.97152 Mbit/s, below200.
2. No. Unchanged window/RTT preserves20.97152 Mbit/s. Remove that limit and other bottlenecks before expecting higher link utilization. Convert bytes to bits;RTT is not one-way delay.

OS17-E04 • DNS cache and identity

1. At t=150 the cached .20 may be returned with 150 s of TTL. At t=300 it expires; because stale use is disabled and the question guarantees a successful query with no later authority change, the resolver obtains .30. A failed query alone would not establish that value. Delegation is by zone, not necessarily one server per label.
2. DNS names identify records; IP addresses participate in routing and ports in transport demultiplexing. Neither is a user or PID.

OS17-E05 • Routes and packet pipelines

1. Initially B=∞; settling D at 1 yields B=6, then settling C at 2 yields B=4. A–C–B costs 4; after removing C–B, A–D–B costs 6.
2. Each packet takes $1000\times8/10^6=8$ ms per hop. Pipeline completion is $(2+4-1)\times8=40$ ms. The whole message takes $4000\times8/10^6=32$ ms per hop, or 64 ms total. These times rely on the stated idle FIFO store-and-forward model.

OS17-E06 • IPv4, ARP, and a short frame

1. A selects next-hop 192.0.2.1 and ARPs for R_L. Frame MAC endpoints are A→R_L then R_R→B; R resolves B on its right LAN. IP endpoints remain 192.0.2.10→198.51.100.20. R decrements TTL and updates the IPv4 header checksum.
2. The IP packet is 20+20+4=44 B; Ethernet data must be at least 46 B, so pad 2 B. The MAC frame is 14+46+4=64 B. Adding 8 B preamble/SFD and 12 IFG byte-times gives 84 byte-times, or $84\times8/10^8=6.72$ µs. IFG is idle time, not IP data.

OS17-E07 • Indistinguishable failure observations

1. In one run B crashes before the ping. In another B remains alive but the request or reply is delayed beyond 200 ms. Through 200 ms A observes the same send and timeout without reply; it cannot infer which run occurred.
2. The earlier heartbeat can arrive at its nominal time; the next can arrive 0.1+0.05 s after its nominal time. The maximum gap is 0.5+0.1+0.05=0.65 s. Loss, partition, exceeded delay bounds, or violated clock assumptions invalidate this no-false-suspicion bound.

OS17-E08 • Fencing and retries

1. A can still write during the partition, so B's self-election can create two writers. After the coordinator issues 43, storage orders epoch installation with each write's validation and commit. A write with 42 linearized after installation is rejected; one linearized before installation was already committed.
2. Commit counter 10→11 and id17→11 in the same durable transaction. A retry finds the retained ID and returns 11. If counter is persisted before the ID, a crash can permit a duplicate increment; ID before effect can lose the increment. At-most-once does not guarantee client knowledge or cover external effects outside the transaction.

OS17-E09 • DFS naming and cache cost

1. Update the vol9 location to serverB while preserving path and existing FID. Reusing file2 requires a new generation so an old FID does not denote the new object.
2. Serial delay is $20\times6+80\times0.2=136$ ms and payload is $20\times4=80$ KiB. Validation adds $100\times1=100$ ms, giving 236 ms.

OS17-E10 • Consistency and partitions

1. Yes. The periodic model permits cached v0 at t=3, so it does not guarantee every new read sees completed v1.
2. Before completing the write, the server needs an acknowledgment that each conflicting read right was actually recalled or must wait for controlled leases to expire safely. The latter requires bounded clock error, defined expiry, and clients that stop using expired cached rights. An unacknowledged invalidation cannot prove a partitioned client stopped. Previously delivered application bytes do not disappear.

OS17-E11 • NFSv3 verifier changes

1. $Y\neq X$ means the old UNSTABLE response cannot establish that D survived. Retain and retransmit bytes not known stable; a partial WRITE requires continuing from its returned count. Once the required range is fully covered, either successful COMMIT with the appropriate matching verifier for the unstable writes, or successful stable WRITEs at the required stability level for each covered byte, releases the corresponding retransmission obligation. DATA_SYNC and FILE_SYNC differ in metadata guarantees.
2. No. NFSv3 basic calls carry filehandle and offset without equivalent OPEN session state, but servers can cache, and WRITE can be UNSTABLE. Durability, visibility and deduplication are distinct.

OS17-E12 • Historical HDFS and failover capacity

1. Block payloads are 64, 64 and 22 MiB, totaling 150 MiB. Three completed replicas give 450 MiB payload, excluding metadata, checksums and allocation overhead. The NameNode supplies namespace and locations; client payload travels directly to or from DataNodes. A Secondary NameNode checkpoints rather than automatically acting as hot standby.
2. In the independent request-server model, one remaining server supplies 100 requests/s against 150/s of load, so at least 50/s cannot be sustained in steady state. Takeover alone does not preserve throughput or SLO.

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz et al., Operating System Concepts, 9th ed., Ch. 17. RFC 1034, RFC 826, RFC 1813. Hadoop 1.2.1 HDFS. OpenAFS.

Original educational synthesis by Lin Cing, licensed under CC BY 4.0.
https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study